CS 310: Algorithms *Mudassir Shabbir, April 20, 2026*

Student ID 1 & 2:*(Do this in pairs)*

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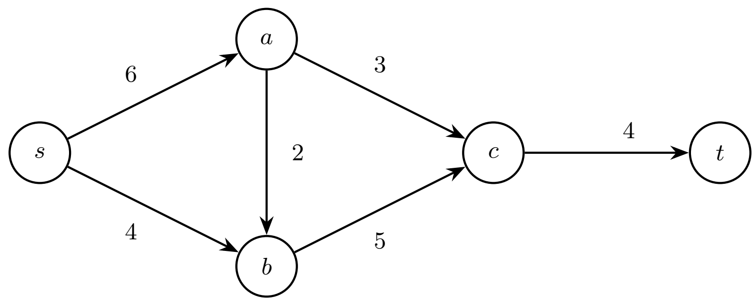
1 Q1: Residual Graph + Flow Rerouting Challenge

Consider the flow network below. Each edge is labeled with its capacity.



1. Perform one Ford-Fulkerson augmentation using the path:

$$s \to a \to c \to t$$

   Find:

   - the bottleneck of this path
   - the updated flow on each edge after augmentation

2. Draw the residual graph after this augmentation step.

3. Find a second augmenting path that uses either:

   - edge $a \to b$, or
   - a backward (residual) edge

   Compute its bottleneck.

4. Compute the final maximum flow value after both augmentations.